

One-Time Pad Version of Vigenere Cipher

14. Write a C program for one-time pad version of the Vigenère cipher. In this scheme, the key is a stream of random numbers between 0 and 26. For example, if the key is 3 19 5 . . . , then the first letter of plaintext is encrypted with a shift of 3 letters, the second with a shift of 19 letters, the third with a shift of 5 letters, and so on.

14(a). Encryption

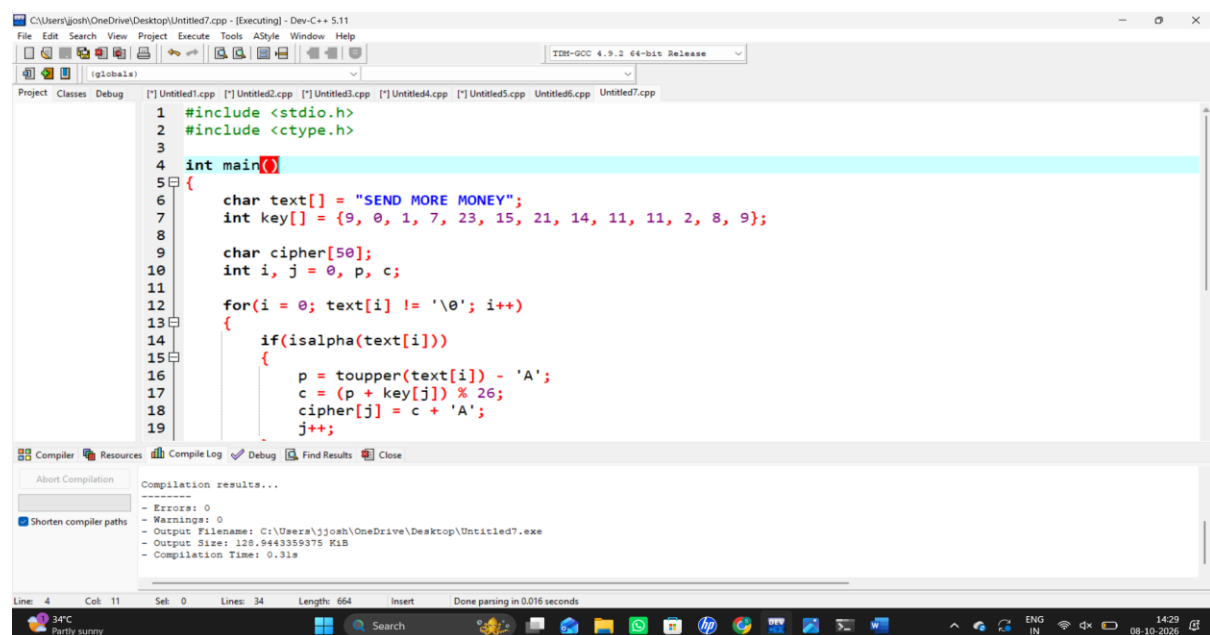
Aim

To encrypt a plaintext using the one-time pad version of the Vigenère cipher with a given key stream.

Algorithm

1. Convert the plaintext into uppercase.
2. Assign A=0, B=1, ..., Z=25.
3. Use the given random key stream.
4. Add the plaintext value and key value.
5. Apply modulo 26.
6. Convert the result into ciphertext letters.
7. Display the ciphertext.

Program

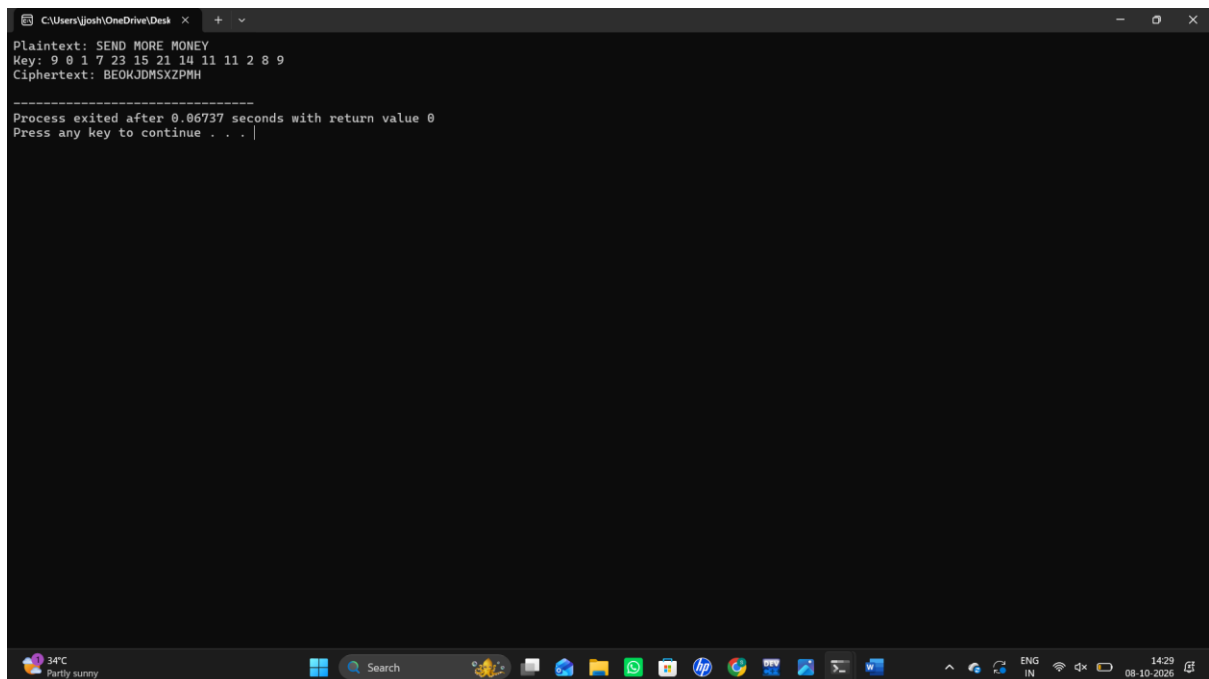


```
1 #include <stdio.h>
2 #include <ctype.h>
3
4 int main()
5 {
6     char text[] = "SEND MORE MONEY";
7     int key[] = {9, 0, 1, 7, 23, 15, 21, 14, 11, 11, 2, 8, 9};
8
9     char cipher[50];
10    int i, j = 0, p, c;
11
12    for(i = 0; text[i] != '\0'; i++)
13    {
14        if(isalpha(text[i]))
15        {
16            p = toupper(text[i]) - 'A';
17            c = (p + key[j]) % 26;
18            cipher[j] = c + 'A';
19            j++;
20        }
21    }
22    printf("Ciphertext: %s", cipher);
23    return 0;
24 }
```

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\josh\OneDrive\Desktop\Untitled7.exe
- Output Size: 128.9443359375 KiB
- Compilation Time: 0.31s

Input/Output



```
C:\Users\josh\OneDrive\Desktop > .\vigenere.py
Plaintext: SEND MORE MONEY
Key: 9 0 1 7 23 15 21 14 11 11 2 8 9
Ciphertext: BEOKJDMXZPMH

-----
Process exited after 0.06737 seconds with return value 0
Press any key to continue . . .
```

Result

Thus, the plaintext was successfully encrypted using the one-time pad version of the Vigenère cipher.

14(b). Find a Key for the Given Plaintext

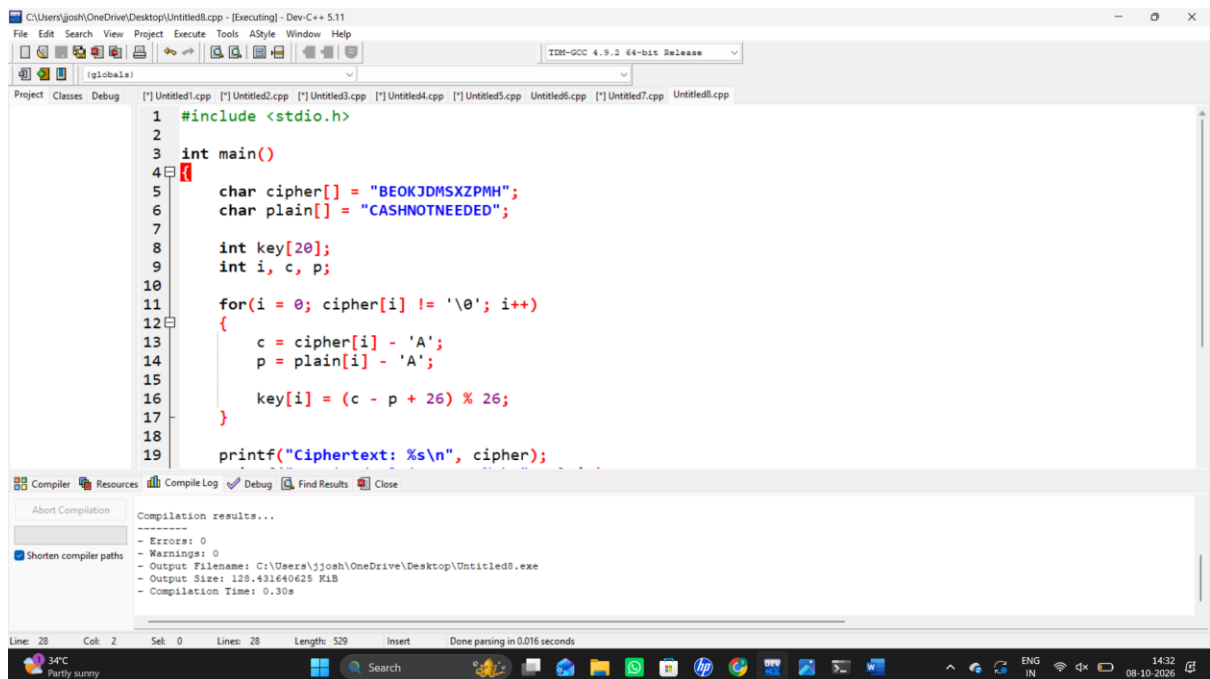
Aim

To find a key stream that decrypts the given ciphertext into the plaintext CASH NOT NEEDED.

Algorithm

1. Take the ciphertext obtained in part (a).
2. Convert ciphertext and desired plaintext into numerical values.
3. Calculate: $K = (C - P) \bmod 26$
4. Repeat for every character.
5. Convert the numerical key values into letters or retain them as numbers.
6. Display the required key stream.

Program

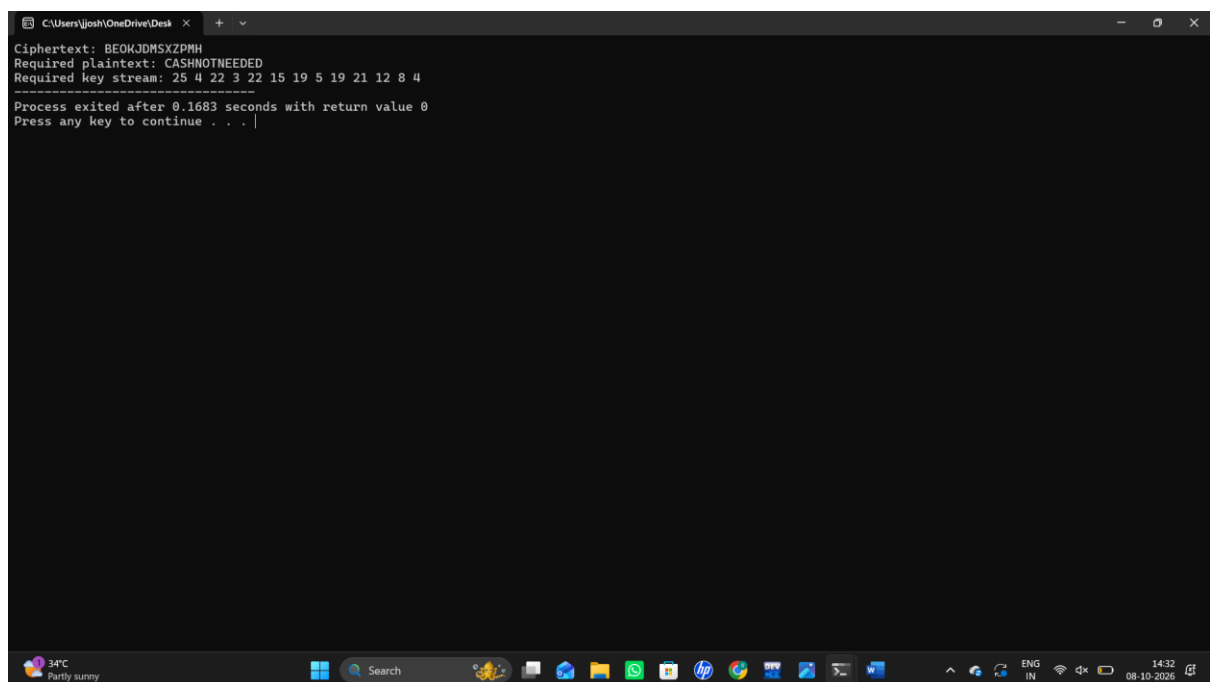


```
1 #include <stdio.h>
2
3 int main()
4 {
5     char cipher[] = "BEOKJDMXSZPMH";
6     char plain[] = "CASHNOTNEEDED";
7
8     int key[26];
9     int i, c, p;
10
11     for(i = 0; cipher[i] != '\0'; i++)
12     {
13         c = cipher[i] - 'A';
14         p = plain[i] - 'A';
15
16         key[i] = (c - p + 26) % 26;
17     }
18
19     printf("Ciphertext: %s\n", cipher);
```

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\josh\OneDrive\Desktop\Untitled8.exe
- Output Size: 128.431640625 KiB
- Compilation Time: 0.30s

Input/Output



```
Ciphertext: BEOKJDMXSZPMH
Required plaintext: CASHNOTNEEDED
Required key stream: 25 4 22 3 22 15 19 5 19 21 12 8 4
-----
Process exited after 0.1683 seconds with return value 0
Press any key to continue . . .
```

Result

Thus, the key stream required to decrypt the ciphertext to CASH NOT NEEDED was successfully obtained.